CHAPTER 5

Role of liver in gallstone formation

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Introduction

Gallstones are lumps in the biliary system or gallbladder that are brought on by unusually high levels of either bilirubin (a byproduct of hemoglobin breakdown) or cholesterol in the bile (Cook et al., 1994). Gallstones are widespread among the general population. The disease affects 10%–20% of adults worldwide, and more than 20% of those with the disease experience symptoms including biliary colic or infections (Everhart & Ruhl, 2009). The symptoms are brought on by the presence of gallstones in the bile ducts or gallbladder. From a clinical standpoint, patients with asymptomatic stones are normally not thought to have gallstone disease and are typically not discovered. Gallstones disorder is one of the digestive disorders with the biggest socioeconomic costs (Shoda et al., 2001).

Epidemiology

Around 20% of adults in Europe suffer from gallstones (Völzke et al., 2005). Since gallstone disease is a chronic disorder that worsens with age and reaches a plateau in prevalence after 50 and 60 years of age in women and men, respectively. The prevalence of this disease in older populations is a good indicator of the disease's worldwide health burden (Miquel et al., 1998). Additionally, women are more likely than men to suffer from gallstones. Hispanic communities in Central and South America, as well as those in the United States with Native American ancestry, have higher rates of gallstones (Everhart et al., 2002). The prevalence of gallstones has historically been lowest in African populations but much higher in Asian countries. Prevalence has

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gradually increased in ethnic groups with previously low prevalence rates as a result of the introduction of high-calorie, high-carb, low-fiber diets, and sedentary lifestyles (Carey & Paigen, 2002; Tsai et al., 2008).

Pathogenesis of gallstones

Gallstones can cause the hepatic, cystic, and common bile ducts to get blocked, among other bile ducts (Ko & Lee, 1999). Bile is transported to the small intestine through the common bile duct from the cystic and hepatic channels. Bile is transported to and from the gallbladder by the cystic duct and from the liver by the hepatic ducts. Bile trapped in these ducts can cause inflammation in the gallbladder, ducts, and liver. One of the ducts that opens into the common bile duct is the pancreatic duct, which runs through the pancreas and carries enzymes that aid in digestion (Donovan, 1999). Pancreatic inflammation can occur from gallstones that travel through the common bile duct. Various mechanisms contribute to the formation of gallstones.

Supersaturation of bile

The equilibrium of cholesterol input and outflow in the liver determines how much biliary cholesterol is secreted (Fig. 5.1). Bile salts and phospholipids cannot solubilize excess cholesterol in bile at equilibrium, which is the case with supersaturated bile (Carey & Small, 1978). A combination of hypersecretion of cholesterol and hyposecretion of bile salts or phospholipids, excessive hepatic cholesterol secretion, decreased hepatic secretion of biliary bile salts or phospholipids with relatively normal cholesterol secretion, or all three of these factors may result in supersaturated bile (Carey & Small, 1978). According to the various relative concentrations of cholesterol, phospholipids, and bile salts, five distinct paths to bile crystallization have been identified. The rates of crystallization in lithogenic human bile are invariably higher than in healthy individuals, regardless of the route (Wang & Carey, 1996). Instantaneous *in vitro* crystallization of cholesterol monohydrate crystals distinguishes supersaturated bile in people without gallstones from lithogenic bile in patients with gallstones (Holzbach et al., 1984; Levy et al., 1984).

Role of mucus

Gallstone development depends on mucus glycoprotein. Gallstone development, biliary sludge production, and the nucleation of bile all depend on the glycoprotein (Lee, 1999). Mucus obstructs gallbladder emptying, impairs

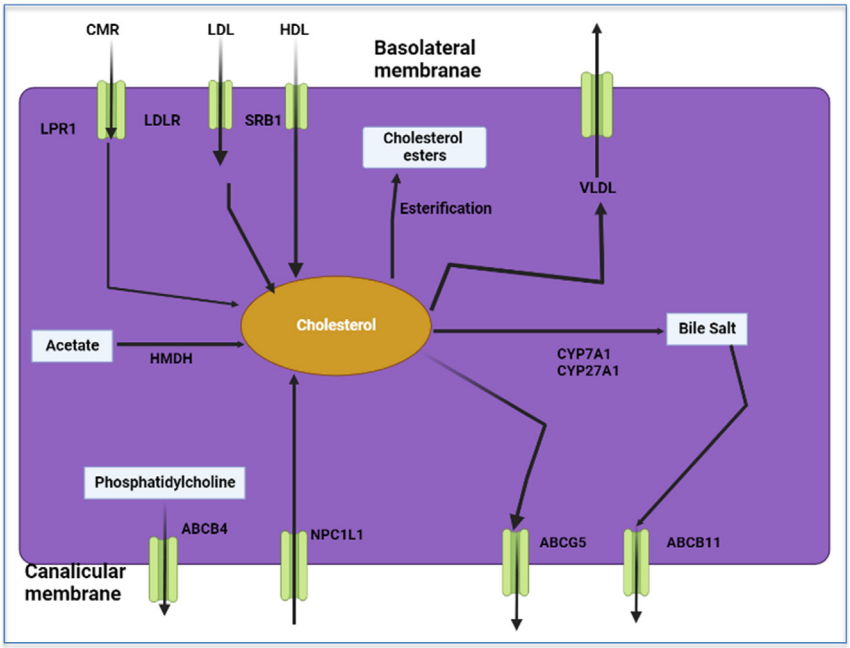

Figure 5.1 ***Cholesterol metabolism in hepatocytes.*** Scavenger receptor class B member 1 (SRB1), low-density lipoprotein receptor-related protein 1 (LRP1), low-density lipoprotein (LDL), and low-density lipoprotein receptor (LDLR) are the mediators of hepatic uptake of cholesterol for chylomicron remnants (CMRs), LDL, and high-density lipoprotein (HDL), respectively. The enzyme 3′-hydroxy-3′-methylglutaryl-coenzyme controls the biosynthesis of hepatic cholesterol from acetate (HMDH). The two rate-limiting enzymes cholesterol 7-hydroxylase (CYP7A1) and sterol 27-hydroxylase (CYP27A1), respectively, control the two pathways that consume a significant amount of cholesterol for the formation of bile salts. Acyl-coenzyme (ACAT) esterifies cholesterol so that it can be stored in hepatocytes. Very-low-density lipoprotein (VLDL), which is secreted into the bloodstream, is made in part from cholesterol. A family of ATP-binding cassette (ABC) transporters, including the heterodimers ABCG5 and ABCG8 for cholesterol, ABCB11 for bile salts, and ABCB4 for phosphatidylcholine, are present in the canalicular membrane and control the secretion of hepatic lipids. The Niemann-Pick C1-like protein 1 (NPC1L1) of the canalicular membrane may facilitate the absorption of cholesterol from hepatic bile into hepatocytes.

the bile salt pool's equilibration, binds bile lipids and bilirubin, and stimulates the growth of cholesterol monohydrate crystals (Ko & Lee, 1999).

Calcium in bile

Cholesterol and pigment stones are formed with the help of calcium. The core of cholesterol stones is formed by calcium bilirubinate, while the

surface calcification of cholesterol stones is caused by the precipitation of calcium carbonate. Calcium facilitates the formation of cholesterol/phospholipid vesicle-derived monohydrate crystals. Normal biliary pH levels prevent calcium from precipitating because they decrease the solubility of calcium products, which prevents the crystallization of calcium carbonate (Taylor et al., 1995). When the gallbladder fails to acidify the bile, precipitation of calcium carbonate and phosphate might happen (Williamson & Percy-Robb, 1980).

Phospholipids in bile

Most of the phospholipids in bile are lecithins. Arachidonic acid phospholipids, which are naturally in excess in patients who develop cholesterol stones, can be converted by the gallbladder mucosa into prostaglandins that increase the production of mucus and decrease gallbladder motility (Kaminski, 1989). This process is thought to be active in people who are predisposed to developing cholesterol gallstones.

Types of gallstones

Gallstones are mainly of three types: cholesterol, pigment, and mixed stones. According to evidence, 90% of people with cholelithiasis have cholesterol stones (Qiao et al., 2013). The main sources of cholesterol in the human body are dietary intake, enterohepatic circulation, and endogenous synthesis. However, cholesterol cannot be converted in the body to CO_2 and water because human and animal cells lack enzymes that can break down the ring structure of the sterol. Therefore, extra cholesterol must be converted into other chemicals or eliminated through feces to stop a potentially dangerous buildup of cholesterol in the body (Taylor et al., 1995).

Cholesterol gallstones

Cholesterol gallstones are the most typical gallstones, and they are mostly formed of cholesterol. Cholesterol is one of the substances that the liver cells produce in the bile, which appears as a white, crystalline powder (Njeze, 2013). The liver purges extra cholesterol during this process. Bile dissolves fatty cholesterol so that it can be transported through the ducts. If the liver secretes an excessive amount of cholesterol, some of the cholesterol does not break down completely. Some of the cholesterol does not dissolve if the liver does not secrete enough bile acids and lecithin.

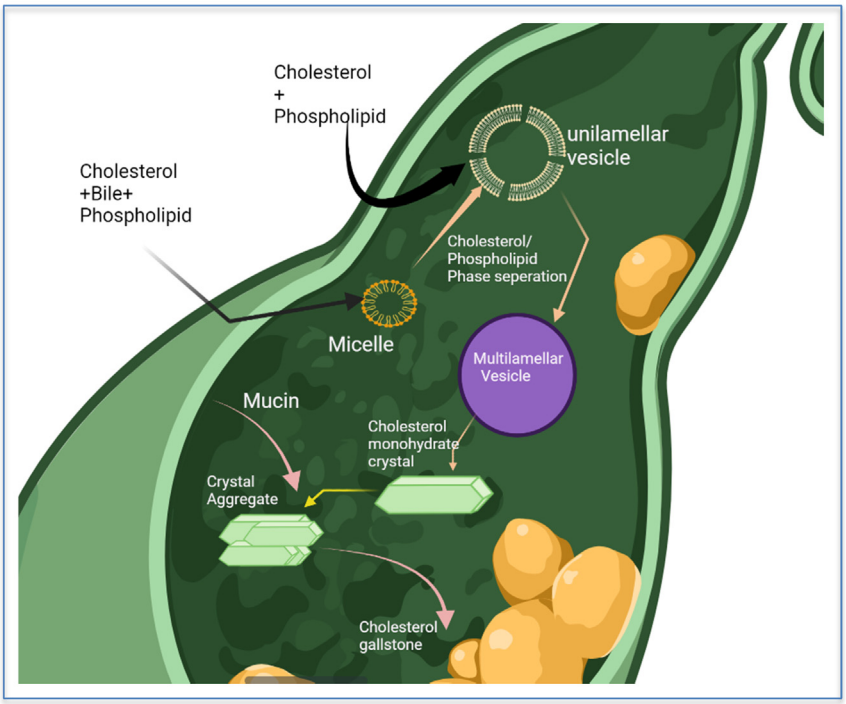

Figure 5.2 ***Schematic representation of cholesterol gallstones arising in Gallbladder.*** Bile salts, cholesterol, and phospholipids interact hydrophobically to generate micelles, cholesterol and phospholipids interact hydrophobically to form unilamellar vesicles. Typically, micelles would be smaller than unilamellar vesicles. The real supersaturated condition arises momentarily as the cholesterol concentration in gallbladder bile rises, primarily due to hepatic hypersecretion of cholesterol. Typically, supersaturated bile indicates that extra cholesterol has phase separated from micelles and formed unilamellar vesicles with biliary phospholipids. In the most usual nucleation process, unilamellar vesicles fuse to produce multilamellar vesicles, which are followed by liquid crystals. These crystals are joined by mucin gel to create cholesterol gallstones.

In both situations, the undissolved cholesterol clings together to produce cholesterol particles that enlarge and eventually crystallize into larger gallstones (Reshetnyak, 2012). Fig. 5.2 provides a schematic representation of how cholesterol gallstones arise.

Pigment gallstones

These stones appear as brown or black stones made of calcium salts of bilirubin (such as calcium bilirubinate) (Bouchier, 1992). Black stones are

tough, primarily found in the gallbladder, made of bilirubin polymers, and free of germs or cholesterol. On the other hand, brownstones are soft and can occur in the gallbladder and the bile duct. The calcium bilirubinate crystals that make them up are radiolucent. These stones typically contain crystals of cholesterol and bacilli-infected bile (Athanasiadou et al., 2013). A very distinctive component of brown pigment stones is calcium palmitate, which is created by the hydrolysis of lecithin.

Black pigment gallstones

A change in pH, a rise in ionized calcium, and a rise in unconjugated bilirubin are all contributing factors. Since infection is not thought to be a factor, it may be possible that aberrant glucuronidase activity on the mucosal surface of the gallbladder is crucial. One recognized causative factor is hemolysis. The formation of black pigment gallstones is described in Fig. 5.3.

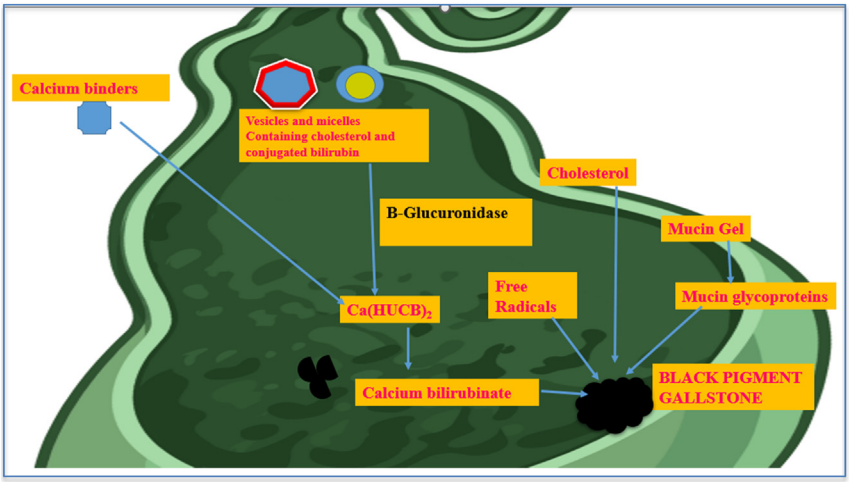

Figure 5.3 ***Formation of black pigment stones in gallbladder.*** Excessive cholesterol in the gallbladder leads to the conjugation of bile containing bilirubin with cholesterol leading to the formation of vesicles and micelles. The micelles are acted upon by beta-glucuronidase mainly of hepatic origin to form monoacidic unconjugated bilirubin (HUCB −), which conjugates with calcium ions. The HUCB − leads to the release of free radicals and also turns to calcium bilirubinate. Mucin glycoproteins along with the other factors finally lead to the precipitation of black pigment stones.

Brown pigment gallstones

The combination of stasis and infection is the main cause of brown pigment stones. Sphincters of Oddi disorders, foreign objects in the ducts, and biliary strictures cause stasis. The main cause of infection is *Escherichia coli*, which causes bilirubin to be hydrolyzed into unconjugated bilirubin; the production of lysolecithin, which helps to generate cholesterol precipitates; and the release of free bile acids, which can also result in the development of calcium salts. Bacteria also make glycoprotein, and they may even increase the production of glycoprotein from the gallbladder, which helps to promote the formation of biliary sludge and the nucleation of crystals of bilirubinate and cholesterol (Thistle et al., 1984) (Fig. 5.4).

Mixed gallstones

These are a mixture of pigment and cholesterol stones made up of different ratios of cholesterol and bilirubinate salts (Shiffman et al., 1991). They also contain other ingredients such as calcium phosphate, calcium carbonate, calcium bilirubinate, bilirubin, calcium palmitate, and calcium stearate. These frequently show up on radiographs because of their high calcium concentration (Francesco Attili et al., 1995).

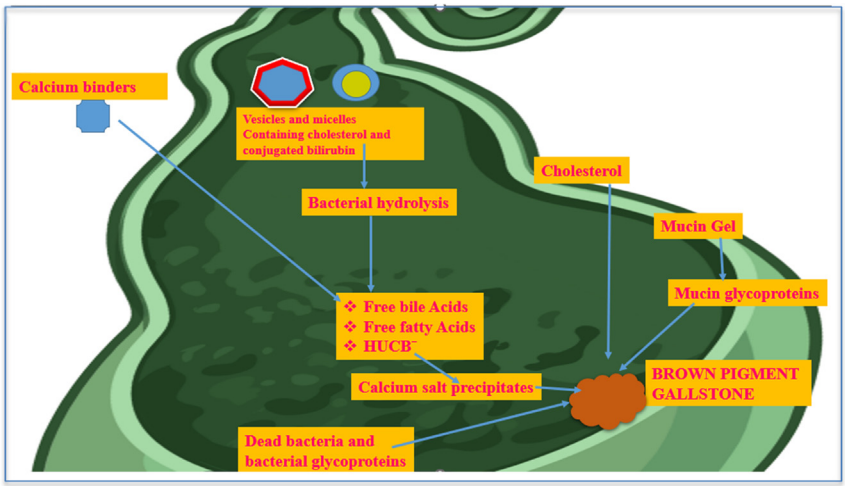

Figure 5.4 ***Formation of brown pigment stones.*** Infection in the gallbladder by bacteria like *Escherichia coli*.

Classification by clinical aspects

Asymptomatic gallstones

Certain gallstones are typically asymptomatic throughout a patient's lifetime. If biliary pain is absent, gallstones are asymptomatic, even when dyspeptic symptoms are present. When abdominal ultrasonography is done for a variety of reasons, it is frequently found by accident (Francesco Attili et al., 1995). According to epidemiological research, gallstones are asymptomatic in between 70% and 80% of the general population, and between 20% and 30% of those gallstones will eventually show symptoms and consequences between five and twenty years following the diagnosis (Maggi et al., 1997). Gallstones are frequently asymptomatic in people with liver cirrhosis and are more likely to be found during routine ultrasonographic liver disease checks (Castaing et al., 1983).

Symptomatic gallstones

Symptomatic gallstones are characterized by pain that might be constant and steady. The most straightforward definition of biliary pain is one that radiates to the back, originates in the right hypochondrium or epigastrium, occurs postprandially (but not necessarily), is severe, and lasts for more than 15–30 minutes (Venneman et al., 2006). Patients with symptomatic gallstones need to be treated with medical dissolution agents targeting the stones, cholecystectomy, and extracorporeal shockwave lithotripsy. Among all the treatments employed, cholecystectomy is a golden measure. Medical dissolution is not regarded as an effective substitute for cholecystectomy due to the low effectiveness and high rates of gallstone recurrence after bile acid dissolution. Randomized controlled trials showed that cholecystectomy was still a better option, although extracorporeal shockwave lithotripsy was effective (Schmidt et al., 2011; Venneman et al., 2006).

Factors affecting gallstone formation

Gallstone formation can be caused by problems related to the gallbladder or other components of the bile-secretion system (Jaruvongvanich et al., 2016). Due to the complexity of the bile-secretion system, there are numerous causes for gallstone development. Gallstones may be linked to age, gender, the physiology of women, obesity, cardiovascular disease, the microbiome, or sugar metabolism. The interaction of all the above factors

was hypothesized based on numerous mice and human research. Hereditary factors, excessive liver cholesterol secretion, impaired motility of the gallbladder, intestinal factors all contribute to the pathophysiology of cholesterol gallstone formation. Intestinal factors can also be divided into two groups, including intestinal microbiota and increased liver-to-small intestine transfer of cholesterol, which ultimately leads to increased bile output. These elements can boost cholesterol crystal growth or manufacturing, which will finally result in the development of stones (Grigor'eva & Romanova, 2020).

Genetic factors for gallstone formation

Mice that are prone to gallstones (C57L/J) and mice that are resistant to gallstones (AKR/J) were used to better understand the genetic components of gallstone development. It was suggested that the lithogenic genes 1 and 2 (Lith1 and Lith2) may contribute to the development of gallstones in these mice. The mouse chromosome 2 gene Lith1 is crucial in determining whether the liver secretes too much cholesterol (Wang et al., 2018). Lith2, found on mouse chromosome 19, controls the bile flow that is dependent on bile salts. The human equivalents of mouse Lith1 and Lith2 are ABCG5 and ABCG8, respectively. ABCG5 and ABCG8 are ABC transporters that are highly expressed in the intestine and hepatocytes (Wang et al., 2015). They either cause the active efflux of cholesterol from the enterocyte back into the intestinal lumen for fecal excretion or they transport neutral sterols to bile in hepatocytes. It was found that the overexpression of the receptors leads to an increase in the cholesterol content in the gallbladder, ultimately leading to gallstone formation (Berge et al., 2000; Wang, Liu, et al., 2020). Studies carried out on patients suffering from gallstones have shown the involvement of the above genes and their variants, and their roles are listed below in Table 5.1.

Gallbladder contraction

Fatty foodstuffs lead to the release of cholecystokinin from the endocrine cells present in the duodenum, which reaches the gallbladder and binds with their receptors in the smooth muscle cells present in the gallbladder, leading to the contraction of the gallbladder and the release of bile into the intestine (Lee et al., 2001). The cholecystokinin receptors are also present in the pancreas, sphincter of Oddi, and gastric mucosa.

Table 5.1 Various genes and their relationship with gallstone formation.

Gene	Variant	Function	References
ABCB4	p.G622E	Codes for cholesterol Transporter ABCB4	Rebholz et al. (2018)
ABCG8	p.D19H rs4953023 rs4299376	Codes for cholesterol Transporter ABCG8	Rebholz et al. (2018)
ABCG5	rs6544718 rs6720173	Codes for cholesterol Transporter ABCG5	Rebholz et al. (2018)
CYP7A1	rs6471717	Codes for enzyme 7α-hydroxylase required for metabolism of bile salts	Rebholz et al. (2018)
GCKR	p.P446L rs1260326	Regulates glucokinase and hyperglycemia	Rebholz et al. (2018)

Cholecystokinin plays an important role in gastric emptying and in regulating pancreatic release; hence, its function is very important regarding gallstone formation. In an experiment on knockout mice where the genes expressing cholecystokinin receptors were knocked out, there was an increase in gallstone formation due to the inhibition of gastric emptying and cholesterol metabolism (Rebholz et al., 2018; Wang et al., 2019).

Microbiome

Gallstone formation is closely related to the microbial flora in the human body because, in all stages of bile production, there is the involvement of the gastrointestinal microbiota. Microorganisms enter the duodenum through the sphincter of Oddi; they also reach the liver through the bloodstream and ultimately into the bile. The microbes play an important role in the nucleation process leading to the formation of pigment gallstones (Miller et al., 2020). Gallstone production can be regulated by the characteristics of the bacteria that live in the gallbladder. While bacteria that create irregularities in mucus are more likely to result in the creation of cholesterol stones. Bacteria that produce beta-glucuronidase and phospholipase in the bile yield a higher amount of pigment in the stones. It was found in various studies that beneficial bacteria like *Roseburia intestinalis* are diminished in GSD patients, and an overgrowth of bacteria of the *Proteobacteria* type—including a variety of dangerous pathogens including *E. coli*, *Salmonella*, *Vibrio*, and *Helicobacter*—occurs more frequently (Wang, Portincasa, et al., 2020).

Role of estrogen in gallstone formation

In various studies conducted, it was found that women have a higher tendency to form gallstones as compared to men, and in pregnant women, the incidences are the highest (Grigor'eva & Romanova, 2020; Helaly et al., 2014; Neri et al., 2005). 17β-Estradiol is a major female steroid hormone that plays a significant role in female reproductive health. It also plays an important role in gallstone formation. The hormone, being a lipophilic substance, easily crosses the cell membrane and binds with the intracellular receptors (ESR1 and ESR2), therefore directly controlling transcription and translation of the required proteins. In the liver cells, 17β-estradiol stimulates the production of excess bile and cholesterol, leading to increased risks of gallstone formation. Clinically, it was found that in pre-menopausal women, the increased usage of estrogen-based contraceptives increased the chances of the formation of gallstones. Estrogens promote the synthesis of cholesterol in some specific pathways in liver cells. The stimulation of sterol regulatory element binding protein-2 leads to an increase in cholesterol production when there is excess estrogen in the bloodstream. Intracellular ESR-1 receptors, which are activated by estrogen, activate ABCG5 and ABCG8, which are expressed in the endoplasmic reticulum of hepatocytes and intestinal cells and play an important role in neutral sterol transportation in the intestinal lumen, leading to bile release and the formation of gallstones (Celaj & Kourkoumpetis, 2021).

Obesity

Gallstone production was discovered to be significantly influenced by nonalcoholic fatty illnesses. Low expression of aquaporin-8 (AQP8), which is mediated by hypoxia inducible factor-1 (HIF-1), is the primary cause of gallstone development. The expression of genes involved in oxygen transport, cell growth, and redox balance is regulated by the essential transcription factor HIF-1. To increase cell survival in hypoxic settings, it promotes an adaptive response (Schwulst & Son, 2021; Wang et al., 2006). Most liver HIF-1α is found in the region near the hepatic vein. Lipid buildup considerably expands hepatocytes throughout the progression of hepatic steatosis, decreasing hepatic sinusoidal perfusion and microcirculation, and finally causing liver hypoxia (De Bari et al., 2015). Researchers found a significant 35% increase in bile flow, a 36% reduction in the concentration of bile lipids in the gallbladder and hepatobiliary

juice, and a decrease in gallbladder inflammation in one study that looked at the upregulated expression of AQP8, a water channel protein that is responsible for the secretion of liver water into the bile duct (Balamurugan, 2016; Karimi et al., 2018; Kondo et al., 2010; Waypa & Schumacker, 2019). Therefore, in the experiments with knockout mice where liver-specific HIF-1α is knocked out, the production of cholesterol crystals was suppressed. On the other hand, it was demonstrated that in wild-type mice with diet-induced fatty livers, activation of the HIF-1 pathway suppressed the development of gallstones. Additionally, the elevated expression of HIF-1 and its downstream targets in the liver raises the possibility that HIF-1 may be crucial in the development of cholesterol gallstones in NAFLD patients (Calamita et al., 2005).

Carbohydrate metabolism

The traditional function of bile in the breakdown and absorption of fat is well known. Furthermore, the equilibrium of glucose, fat, and energy is regulated physiologically by the gallbladder. Insulin sensitivity can be decreased by both GSD and cholecystectomy (Arrese et al., 2018), indicating that fat is a risk factor for both IR and gallstones rather than being correlated with either condition. Triglyceride levels in the liver are raised by GSD and cholecystectomy, and it is possible that both procedures also raise IR levels. However, in addition to controlling bile acid secretion and transport, the gallbladder also influences lipid and glucose homeostasis, which may have an impact on the body's overall energy expenditure (Arrese et al., 2018; Nervi & Arrese, 2013).

Treatment

Observation for asymptomatic gallstones

Patients with asymptomatic gallstones must be first put under surveillance to observe the symptoms because the surgical procedure is expensive and painful, with some chances of patient mortality. The same procedure of symptom observation and later surgery must also be followed for cirrhotic patients with silent stones.

Prophylactic cholecystectomy

The surgical removal of the gallbladder and gallstones is known as a cholecystectomy. Cholecystectomy can be divided into two types: prophylactic

and laparoscopic (Castaing et al., 1983; Stender et al., 2014). For gallstone carriers in the general population, prophylactic cholecystectomy is typically not advised, with a few exceptions. Due to their higher surgical risk compared to people without liver disease, it should also not be advised for cirrhotic patients. It is debatable how to treat asymptomatic gallstones that were unintentionally discovered in these people while having abdominal surgery for another reason. Patients with well-compensated cirrhosis who are having elective abdominal surgery for another ailment may be a good candidate for concurrent cholecystectomy (Gurusamy & Samraj, 2007).

Spectroscopic approaches to analyze and detect gallstones

The analysis of gallstones and kidney stones is done using cutting-edge multi-spectroscopic methods like Fourier transform infrared (FTIR), ultraviolet-visible, wavelength dispersive X-ray fluorescence (WDXRF), laser-induced breakdown spectroscopy (LIBS), and time-of-flight secondary ion mass spectroscopy (TOF-SIMS). Following is a brief explanation of the technique and methodology of these procedures, and their use and applications in gastroenterology and urology are discussed (Ishizaki et al., 1993).

Fourier transform infrared

Through the measurement of IR absorption spectra, FTIR spectroscopy is a potent method for classifying substances into organic and inorganic parts. It may offer in-depth knowledge of the composition of the intended item. Quantitative analyses can also be done using pertinent standards. Gallstones and kidney stones can be analyzed using FTIR, which can also evaluate a variety of other objects. The quantity of molecules in the target sample is shown by the reporting of specific vibration band absorption in FTIR-based spectral analysis of stones in the frequency ranges $4000-400\ cm^{-1}$. It is simple, quick, accurate, and nondestructive to perform FTIR spectroscopy measurements in nature. Numerous studies of the structure and conformation of proteins, fatty acids, carbohydrates, and nucleic acids have been conducted using them (Ha & Park, 2018; Raman & Selvaraju, 2008; Singh & Rai, 2014).

Wavelength dispersive X-ray fluorescence spectroscopy

Due to its accuracy, precision, sensitivity, and dynamic linear range, X-ray fluorescence is a helpful analytical tool for the investigation of the

elemental composition of bulk materials, including biomaterials. It can handle a wide range of weight fractions, from traces to pure elements, and it covers a large range of components. It offers elemental analysis based on the measurement of distinctive X-rays that the sample emits. Characteristic X-rays are released from the atoms of the target material as a result of the powerful X-ray bombardment, and these can be utilized to determine the sample's elemental makeup. One of the two broad groups of X-rays is WDXRF. Fluorescence techniques are used to measure elements, and they are very helpful for biomedical applications. All of the components of the sample are simultaneously energized in the WDXRF spectrometer during WDXRF spectroscopy. An analyzing crystal will then diffract the various energies of the characteristic X-ray emission from the target materials in various directions. Thus, using a detector at a specific angle, the intensities of X-rays emitting at a given wavelength are detected (Auti et al., 2022; Jaswal et al., 2016; Makhal et al., 2021; Singh et al., 2017).

Time-of-flight secondary ion mass spectrometry

For both inorganic and organic materials, TOF-SIMS is a valuable surface analysis method. In TOF-SIMS, secondary chemical ions are typically removed from the surface of the target materials using a pulsed particle beam made up of either Bin + , Cs + , Ga + , or Ar + . The desorption of secondary positive and negative ions as well as electrons from the target material's surface follows. After being driven through a flight tube, these surface-derived secondary ions are measured for mass by calculating the precise moment at which they arrived at the detector. Depending on the characteristics of the primary ion beam, TOF-SIMS can also reveal information on the target sample's structural and molecular characteristics, in addition to information about its elements. It is currently used to examine different types of kidney stones and gallstones (Chattaraj et al., 2022; Ghumman et al., 2012; Jaswal et al., 2019).

Laser-induced breakdown spectroscopy

Solid, liquid, and gaseous materials, including biological species, can be analyzed using LIBS, an optically sensitive laser-based elemental analysis technology. This method relies on the sample being exposed to a high-power pulsed laser beam, which causes a short-lived laser-induced microplasma to develop. The molecules from the target materials are

broken down into atomic and ionic forms, and the optical emission from the plasma gives information on the chemical makeup of the target sample. The collecting optics (often an optical fiber bundle) were used to collect the plasma created by the laser and feed it into the high-resolution spectrometer. The vital information about the elements and their quantities in the sample was thus disclosed by the LIBS spectra. The atomic emission approach is a very straightforward, quick, and reliable way to examine kidney and gallstones (Chowdhury et al., 2020; Gazali et al., 2021; Patel et al., 2021; Singh et al., 2008, 2009; Sodhi et al., 2011).

Conclusion

Gallstone disease is spreading at an alarming rate, breaking all geographical borders, and lifestyle changes are playing an important role in the spread. The liver plays an important role in gallstone formation using various physicochemical mechanisms operational in the human body. The current treatments are sufficient to treat the disease but it's important to understand the pathophysiology so that the right and timely treatment can be provided to the patient.

Conflicts of interest

The authors declare that they have no conflicts of interest.

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